AS-P1-RPT-001 Technical Brief

Rev 1

Maintenance Platform Verification

2026-10-21

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Student exercise. Not for construction.

The gravity and lateral framing of a 6.0 m × 3.0 m steel maintenance platform was verified by three independent methods, and the review comments received on R0 were addressed in this revision.

PURPOSE AND SCOPE

The purpose of this brief is to summarize the verification of a steel maintenance platform, which was carried out as a self-directed student exercise to check the member forces, stresses and deflections of the framing by three separate methods and to record the discrepancies found between them. In R0, the checks were limited to elastic strength of materials, which covered the bending stress, shear stress and deflection of the beams and joists and the axial stress and Euler buckling of the posts. In R1, the bracing was then remodelled as tension-only, the lateral loads were checked by hand (CAL-003), and the members were checked to CSA S16 (CAL-004). Apart from the brace end bolt, connections were not designed, and for this reason the work is not intended for construction.

DESIGN BASIS

Item	Basis
Geometry	6.0 m × 3.0 m plan, top of steel 3000 mm
B1 main beams	W310×39, simply supported, 6.0 m span
J1 joists	W150×22 at 1.5 m (5 No.), 3.0 m span
C1 posts	HSS127×127×6.4, 3.0 m, pinned, K = 1.0
BR1 bracing	L64×64×6.4 X-bracing, all four sides, tension-only, bolted at the intersection
Dead load	0.5 kPa grating + self-weight
Live load	4.8 kPa adopted. The minimum for equipment areas and service rooms is 3.6 kPa (NBC 2020, Div. B, Table 4.1.5.3, as adopted by O. Reg. 163/24), and 4.8 kPa was selected to allow for components and tools laid down on the deck during maintenance (cf. storage areas, 4.8 kPa).
Equipment load	10 kN at midspan of each B1 (assumed)
Notional loads	0.005 × 187.32 kN total ULS gravity = 0.94 kN, applied as 0.234 kN at each post top, in X and Y separately (CSA S16, Clause 8.4.1)
Guard load	0.75 kN/m horizontal, outward on one edge at a time: short edge (X, 3 m) and long edge (Y, 6 m) (NBC 2020, Div. B, Article 4.1.5.14)
Load combinations	ULS 1.25D + 1.5L; ULS with lateral 1.25D + 1.5L + 1.5G + 1.0N, with full floor live load combined with guard load (conservative, for review); SLS D + L and D + L + G
Deflection limit	L/360 under D + L adopted, which is more stringent than the L/300 recommended for floors of industrial buildings in CSA S16, Annex D. For B1, D + L gives 10.72 mm (L/560) and live load only gives 9.43 mm (L/636). Checking D + L against a live load limit is conservative.
Lateral support	B1 compression flange braced by J1 at 1.5 m (L_u = 1.5 m); J1 taken as unbraced over 3.0 m, with no restraint assumed from the grating; ω_2 = 1.0
Material	CSA G40.21 350W (assumed), E = 200 000 MPa; HSS taken as Class C (n = 1.34)
Bolts	Assumed 1 M20 A325 per brace end, threads intercepted, 22 mm hole; F_u = 825 MPa (bolt), 450 MPa (350W plate)

Section properties were taken from the CISC section database (beamdimensions.com), and loads marked as assumed were not taken from a cited source.

VERIFICATION METHOD

The framing was checked by three methods which were set up independently of one another, in order that an error in any one method would be shown as a disagreement with the other two:

1. Hand calculation in Mathcad Prime (AS-P1-CAL-001), carried out member by member from joist to beam to post.
2. Parametric spreadsheet in Excel (AS-P1-CAL-002), in which every input was given as a named cell and a CHECK tab was used to compare all three methods, a difference of 2% or less being taken as a pass.
3. 3D frame model in Autodesk Robot Structural Analysis (AS-P1-MOD-001), with member self-weight included and joist end releases set to free both bending rotations while torsion was restrained.

Additionally, the framing was drawn in AutoCAD (S-101 and S-201) and modelled in Revit (AS-P1-MOD-002, sheet S-301), with the Revit schedules being used for the material take-off and as a further check of the modelled sections. In R1, the brace and post forces and the frame displacements under the lateral load cases were also checked by hand in AS-P1-CAL-003.

RESULTS

All beam and joist results were found to agree within 0.1%, and the only difference above that was the post axial load, which is explained under Findings.

Quantity	Unit	Mathcad	Excel	Robot	Robot vs hand	Limit
J1 factored moment M_f	kN·m	13.51	13.52	13.51	0.0%	
J1 factored shear V_f	kN	18.02	18.02	18.02	0.0%	
J1 service deflection	mm	3.56	3.56	3.56	0.0%	L/360 = 8.33
B1 factored moment M_f	kN·m	78.70	78.70	78.69	0.0%	
B1 factored shear V_f	kN	35.96	35.96	35.95	0.0%	
B1 service deflection	mm	10.72	10.72	10.72	0.0%	L/360 = 16.67
C1 factored axial load	kN	46.03	46.03	46.83	+1.7%	
C1 Euler buckling load P_{cr}	kN	1546	1546	not run		

The elastic stresses obtained by hand were 143 MPa bending and 20 MPa shear in B1, 85 MPa bending and 20 MPa shear in J1, and 16 MPa axial in C1, with the factored post load being about 3% of its Euler load. These values are below F_y = 350 MPa, and the member resistances to CSA S16 are given under R1 revisions. The small Mathcad and Excel differences in J1 are attributed to g being taken as 9.807 and 9.81 m/s^2 respectively.

R1 REVISIONS

Tension-only bracing. The X-bracing was modelled as tension-only in MOD-001 R1. Under gravity load all of the braces are placed in compression, which makes them inactive, and because of this the full gravity load is carried by the posts. The C1 member force from Robot was then 46.55 kN at the post base, which is within 1.1% of the 46.03 kN hand value. The support reaction of 46.83 kN is higher, since the brace self-weight is lumped at the base nodes.

Lateral loads. The notional loads and the guard load were applied as separate lateral load cases in X and Y. Since the braces act in tension only, one brace in each X is active for a given load direction. The brace and post forces were then checked by hand in CAL-003, and these agreed with Robot within 0.2%.

Item	Case	Hand	Robot	Diff.
Brace tension, bars 22/24 (bottom/top)	ULS+GY	5.44 / 5.66 kN	5.43 / 5.66 kN	0.2% / 0.0%
C1 axial force	ULS+GY	50.40 kN	50.40 kN	0.0%
Brace tension, bars 18/20	ULS+GX	2.41 / 2.63 kN	2.41 / 2.63 kN	0.0%
C1 axial force	ULS+GX	47.63 kN	47.63 kN	0.0%

ULS+GY is 1.25D + 1.5L + 1.5GY + 1.0NY, and it governs the post and brace design.

Displacements. Under SLS+GY (D + L + GY, unfactored), the drift at the post top was 0.30 mm (H/10 000), against 0.29 mm by hand. The lateral deflection of B1 at midspan was 4.65 mm (L/1290), against 4.67 mm by hand. The drift under SLS+GX was 0.15 mm.

CSA S16 member checks. The members were checked to CSA S16 in CAL-004, using the governing factored effects from the gravity and lateral load cases. All members pass, and the highest utilization is 0.53 in B1 under ULS+GY. In addition, the B1 result depends on lateral restraint from J1 (Finding 8). The weak-axis moment in B1 was taken as an upper bound, with the factored guard load of 1.125 kN/m carried by B1 alone as a simple span ($wL^2/8$), although in the frame it is shared with the joists and bracing.

Member	Check	S16 clause	Factored effect	Factored resistance	Util.
B1	Bending, Class 2, L_u = 1.5 m	13.6	M_f = 78.70 kN·m	M_r = 192.2 kN·m	0.41
B1	Shear	13.4.1.1	V_f = 35.96 kN	V_r = 373.8 kN	0.10
B1	Biaxial bending, ULS+GY	13.8.3	M_{fx} = 78.70, $M_{fy} \leq$ 5.06 kN·m	M_{rx} = 192.2, M_{ry} = 42.5 kN·m	0.53
J1	Bending, Class 4 flange, L_u = 3.0 m	13.5(c), 13.6	M_f = 13.51 kN·m	M_r = 44.3 kN·m	0.30
J1	Shear	13.4.1.1	V_f = 18.02 kN	V_r = 183.3 kN	0.10
C1	Compression, KL = 3.0 m, KL/r = 61.5	13.3.1	C_f = 50.40 kN	C_r = 661 kN	0.08
BR1	Tension, gross section yield	13.2	T_f = 5.66 kN	T_r = 241.9 kN	0.02
BR1	Net section, shear lag 0.60A_n	12.3.3.3, 13.2	T_f = 5.66 kN	T_r = 124.7 kN	0.05
BR1 bolt	Shear, threads intercepted	13.12.1.2	V_f = 5.66 kN	V_r = 87.1 kN	0.07
BR1 bolt	Bearing on 6.35 mm leg	13.12.1.2	B_f = 5.66 kN	B_r = 137 kN	0.04

In addition, the brace end connection was checked with one assumed M20 A325 bolt, and bolt shear was found to govern. The block shear, edge distances and end distances were not checked, since the connection geometry has not been fixed.

FINDINGS

The bracing was found to govern the review, since its slenderness requires a connection at the brace intersection and a tension-only arrangement. Each finding has been logged in the review register (AS-P1-MTO-001, REVIEW_REGISTER tab).

No.	Item	Finding	Action	Status
1	C1 reaction	Robot gave 46.83 kN against 46.03 kN by hand, since Robot includes the bracing self-weight (2.6 kN in total, about 0.8 kN per post at ULS).	Accepted as explained.	Closed
2	C1 member force	In R0 the Robot member force of 41.9 kN was lower than the hand value, since the bracing carried part of the gravity load.	Braces modelled as tension-only in MOD-001 R1; all braces inactive under gravity, and C1 = 46.55 kN against 46.03 kN by hand (+1.1%).	Closed
3	BR1 slenderness	With r_{min} = 12.5 mm (computed from I_{yp} and A), L/r is about 537 over the full length of the long braces, against limits of 300 in tension and 200 in compression.	Braces bolted at the intersection and treated as tension-only (S-101, S-201), and modelled as tension-only in MOD-001 R1. L/r = 268 (long) and 170 (short) at half length, below 300.	Closed
4	BR1 sag	A self-weight sag of about 24 mm (about L/280) was shown in Robot for the long braces.	Crossing bolt not modelled. In R1 the braces are truss members, so sag is not computed in Robot.	Open
5	Revit model	The schedule volume check showed the braces modelled as W6X15 (area about 2950 mm^2 against about 768 mm^2 expected).	Type corrected to the L64×64×6.4 equivalent and the schedule re-checked.	Closed
6	Material take-off	1298 kg of members was obtained from CISC mass per metre, against 1327 kg from the Revit volumes (+2.3%, imperial family geometry). An assumed 10% connection allowance gives 1428 kg in total.	Accepted, the allowance to be confirmed.	Closed
7	J1 flange class	b/2t = 11.5, which exceeds the Class 3 limit of $200/\sqrt{F_y}$ = 10.7, so the J1 flange is Class 4 in 350W.	Effective flange width used in CAL-004 (Clause 13.5(c)); utilization 0.30.	Closed
8	B1 lateral support	The B1 bending resistance assumes the compression flange is braced by J1 at 1.5 m. Unbraced over 6.0 m, the utilization would be 1.01.	Joist-to-beam connection to provide lateral restraint, to be confirmed in the connection design.	Open

REVIEW COMMENTS AND RESPONSES

Comments on R0 were received from S. Al-Zubaidi, P.Eng. on 2026-10-18, and the responses made in R1 are listed below.

No.	Comment (summary)	R1 response
R1	Model the braces as tension-only, apply lateral loads, and check forces and displacements.	Done in MOD-001 R1 and CAL-003 (see R1 revisions).
R2	Check whether Robot lets any brace carry compression.	Checked. All braces are inactive under gravity, and Finding 2 is closed.
R3	Identify the code provisions and explain the selected criteria.	Live load and deflection criteria cited in the design basis (NBC 2020 Table 4.1.5.3; CSA S16 Annex D).
R4	Include guard loading in the lateral analysis, and check member forces and connections.	Guard load of 0.75 kN/m included, and member forces checked in CAL-003. Brace end bolt checked in CAL-004 (assumed 1 M20 A325; bolt shear and bearing). Other connections not designed.
R5	10% connection allowance acceptable at this stage.	Retained as a preliminary allowance.
R6	Strength checks do not replace CSA S16 resistance checks.	Member checks to CSA S16 added in CAL-004.

DOCUMENTS ISSUED

Document	Description	Rev	Document	Description	Rev
AS-P1-CAL-001	Mathcad hand calculation	R0	AS-P1-MOD-001	Robot 3D model and result plots	R1
AS-P1-CAL-002	Excel calculation and three-way check	R0	S-101, S-201	AutoCAD plan, sections, notes	R3
AS-P1-CAL-003	Hand check of lateral loads, brace forces and displacements	R1	AS-P1-MOD-002, S-301	Revit model, schedules, sheet	R0
AS-P1-CAL-004	CSA S16 member checks	R1	AS-P1-MTO-001	MTO, slenderness, review register	R1